Week 2 Report – Data preparation

What the files mean:

File	What it is	Number of rows
ons_monthly.csv	ONS monthly vacancies, one row for each 3 month period recorded	304 (Apr-Jun 2001 to Jul-Sep 2026)
ons_indeed_aligned.csv	ONS matched to Indeed 3 month averages over the same months	78 (Feb-Apr 2020 to Jul-Sep 2026) – number of overlapped months recorded on ONS and Indeed
Week 2 Chatfield Project.py	Python script that builds on week 1's file and creates the 2 csv files above and aligns the two datasets.	
indeed_gb_downloaded_2026-11-01.csv	Dated copy of the raw Indeed file used.	

Transformations for each dataset:

ONS:

1. Kept only the monthly rows and excluded the rows that show the 3 month periods.
2. Checked for any missing values, duplicate months and skipped months, where none were found
3. Added the monthly log change
4. Added flags for COVID windows and the approximate first publication month.

Indeed:

1. Kept the seasonally adjusted index for the total postings and new postings
2. Checked for missing values and skipped days where none were found.
3. Averaged the daily values within each month and kept only months with at least 28 days of data. This excluded October 2026 since it was unfinished.
4. Took a 3-month rolling average of the monthly averages so that each value matches the same three months as the ONS data periods.

Date alignment

The datasets are joined on window_end which is the last month of each 3 month window. The ONS labels a window as its middle month, so the ONS label is moved forward one month as Indeed's rolling average is already labelled by its last month in the period.

For example:

ONS dataset label	Months covered	window_end	Indeed value used
2026 AUG	Jul-Sept 2026	2026-09	Mean of the July, August and September 2026 monthly averages

Only the windows that are covered in both datasets are kept in the aligned csv. There are 78 of these overlapped periods from Feb-Apr 2020 to Jul-Sept 2026. The 225 ONS windows before Indeed starts are included in the ons_monthly.csv only and these are for the use of ARIMA.

Data Dictionary

ons_montly.csv:

Column name	Meaning	Unit
ons_label_month	The month ONS uses to label the period (the middle month)	YYYY-MM
window_start	First month of the 3-month period	YYYY-MM
window_end	Last month of the 3-month period	YYYY-MM
window	Name of the period	text
vacancies_k	ONS vacancies, seasonally adjusted 3-month average	thousands
dlog_vacancies	100 x change in log vacancies from the previous window	≈ %
covid_window	Whether the period includes any months from March 2020 to Jun 2021	0 (no) or 1 (yes)
provisional	One of the latest 3 windows, which the next ONS release can revise	0 (no) or 1 (yes)
approx_first_published	Month the period was first published (the month after window_end)	YYYY-MM

Ons_indeed_aligned.csv:

Also includes window_end, window_start, window, ons_label_month, covid_window, provisional and approx_first published as shown above.

Column	Meaning	Unit
ons_vacancies_k	Same as vacancies_k	thousands
indeed_total_sa_3m	Indeed total postings (seasonally-adjusted), 3-month average of monthly averages	Index, using 1 Feb 2020 = 100
indeed_new_sa_3m	Same for new postings (on Indeed 7 days or fewer	index
indeed_total_sa_3m_survey_dates	Total postings on the first Friday of each month averaged, averaged over the period	index
dlog_ons	100 x change in long ONS vacancies	≈ %
dlog_indeed_total	100 x change in log indeed_total_sa_3m	≈ %

Limitations

- Publication timing
    - ONS publishes vacancies usually within six weeks of the survey date. In practice this si a month after the window ends. Indeed is about a week behind in real time, so a full window of Indeed data exists before the matching ONS figure is published. Approx_first_published is a rule of thumb not the exact publication date of each window.
- Missing observations
    - Neither source had any missing values or skipped dates. The only data left out on purpose is from the unfinished latest month for Indeed which is October 2026. The first aligned window is Feb-Apr 2020 because the Indeed dataset only starts on 1 Feb 2020.
- Revisions – both files are the latest versions, not what the forecaster saw at the time
    - ONS revises the previous three months in each publication release, as well as more revisions each spring. The latest three windows are flagged provisional.
    - Indeed re-estimates its seasonal adjustment as data arrives and revised its whole history in November 2024 when the method used was changed. A dated copy of the Indeed file is saved each run, so results can be reproduced.

- Overlapping windows
    - Each 3-month average shares two months with the next one, so monthly changes are smoothed and correlated with each other.
- COVID flag
    - Covid_window = 1 if any month between the time frame Mar 2020 to Jun 2021 is included in the window. This gives 61 non-COVID aligned windows from the 78 total as this is using the facts that the value is corresponding to the middle month of the window.
- Survey date
    - ONS counts vacancies on a set date near the start of the month (the first Friday of the month). The first Friday Indeed version approximates this, and the exact ONS count dates were not used.

Sources

- ONS VACS01: Vacancies and unemployment
- ONS Vacancy Survey QMI
- Indeed Hiring Lab job_postings_tracker